AE 358 RENEWABLE ENERGY SYSTEMS AND MANAGEMENT

LECTURE 6

BIOGAS

Lecturer: Francis Kemausuor

COURSE STUDY GUIDE

Date	Meeting/ Week #	Topic for the day	Activity
14-16 Jan	1		Lecture
19-23 Jan	2	Introduction to Energy Systems	Lecture
26-30 Jan		Renewable Energy Systems and their Potential	
2-6 Feb	3	Bioethanol	Lecture
9-13 Feb	4	Hydropower / Group assignments	Lecture
16-20 Feb	5	Biodiesel	Lecture
23-27 Feb	**6**	**Biogas**	**Lecture**
2-6 Mar	7		Lecture
9-13 Mar	8	Mid-semester exams	Quiz
16-20 Mar	9	Discuss group assignment challenges	Lecture
23-27 Mar	10	Solar Energy	Lecture
30 – 3 Apr	11	Wind Energy	Lecture
6-10 Apr	12	Renewable Energy Applications in Agriculture	Lecture
13-17 Apr	13	GROUP PRESENTATIONS	Student-led
13-17 April	14	GROUP PRESENTATIONS	Student-led

Housekeeping

- We are ahead of schedule with AE 358
- Next week period to be used for AE 360 to make up for lost grounds
- Mid-semester examinations on 12 March / 19 March.
- **The Energy Center Seminar Series: Wednesday 4 March, 2015. Biomass assessment for electricity in Ghana – Francis Kemausuor**
- Possibility of a day's lecture on renewable energy in Ghana

Recap

- **Introduction to energy systems**
  - Global energy systems
  - Advantages and disadvantages of fossil fuels
- **Renewable energy sources**
  - Sources available
  - Brief description of their benefits
- **Biofuels**
  - Biodiesel
  - Bioethanol
- **Hydropower**

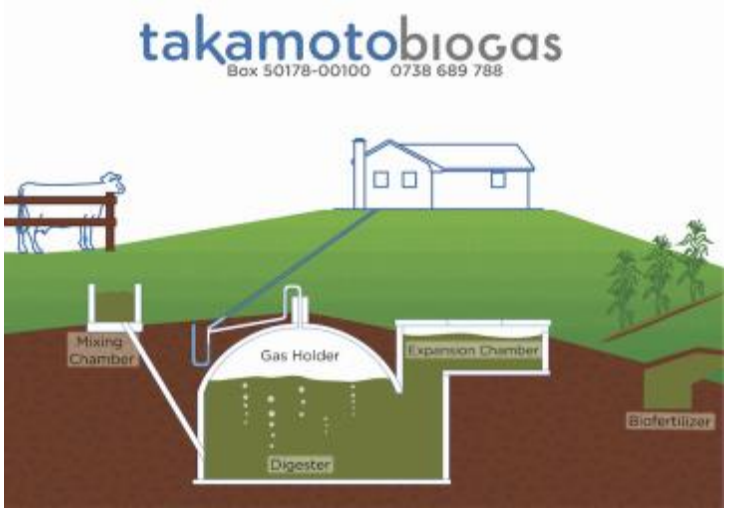
Lecture outline

- What is biogas?
- What are its feedstock sources?
- How is biogas obtained?
- What are the properties of biogas?
- What are the uses of biogas?
- Digester designs
- Construction of biogas plant



What is Biogas?

- Biogas can be defined as a gas composed principally of a mixture of methane and carbon dioxide produced by anaerobic digestion.
- It can be utilized for many purposes, including lighting, heating and power generation at the individual and community levels.
- Biogas technology offers many benefits, including that it can provide sanitation for agriculture through its use of dung (biosolid fraction) and effluent (liquid fraction).





Feedstocks for biogas production



- Slurry or manure (e.g. From dairy farms or “industrial farming” involving large feedlots)
- City sewage and refuse
- Residue from industry - food processing, brewing and distilling, the production of materials such as pulp, paper and pharmaceuticals





Feedstocks for biogas production



- Farming crop residues (e.g. Straw or parts of cereal or fodder plants not normally harvested),
- Direct “fuel crops”, such as ocean−grown algae or seaweeds, water hyacinths (in tropical climates) or fast growing bushes or trees.

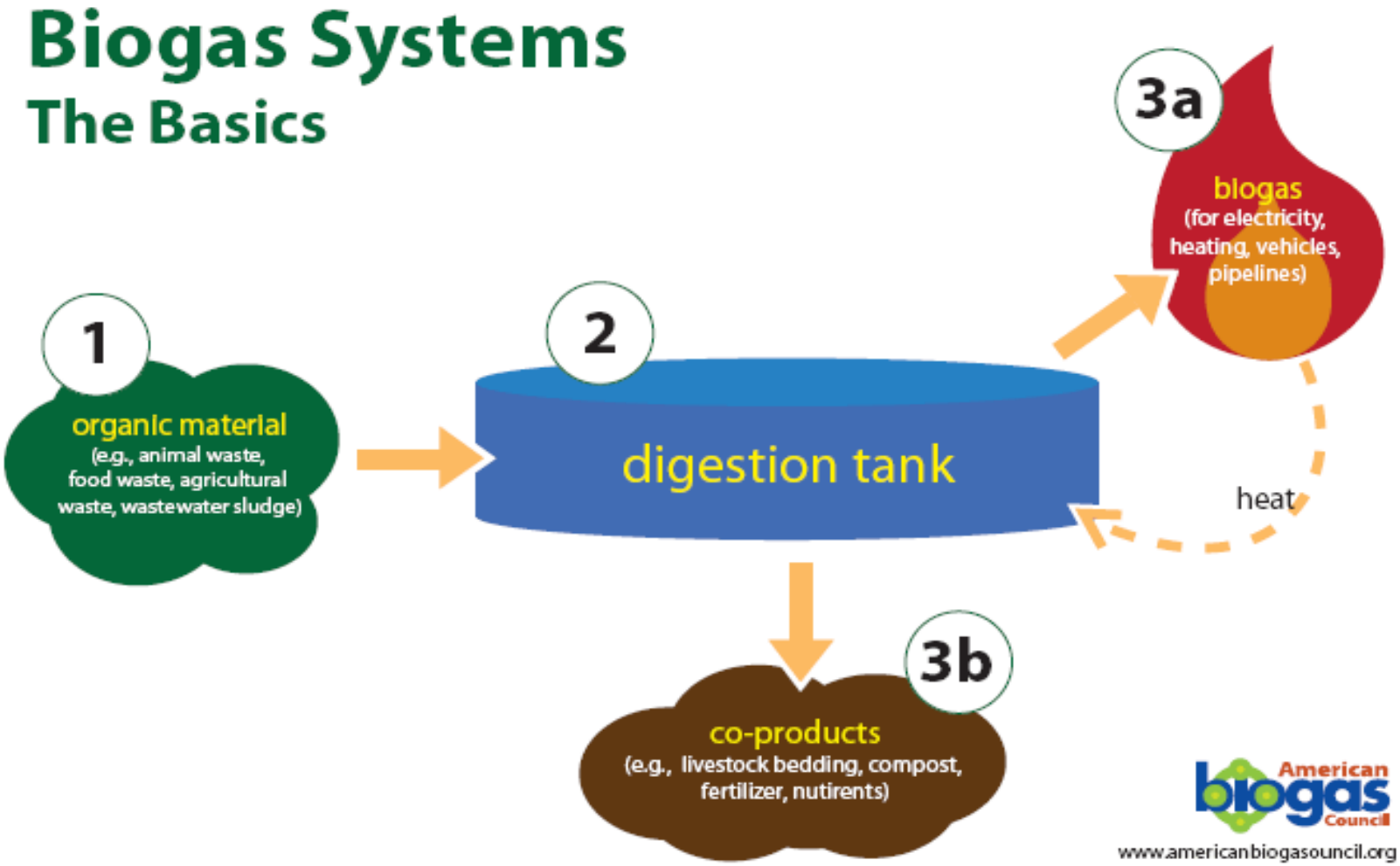




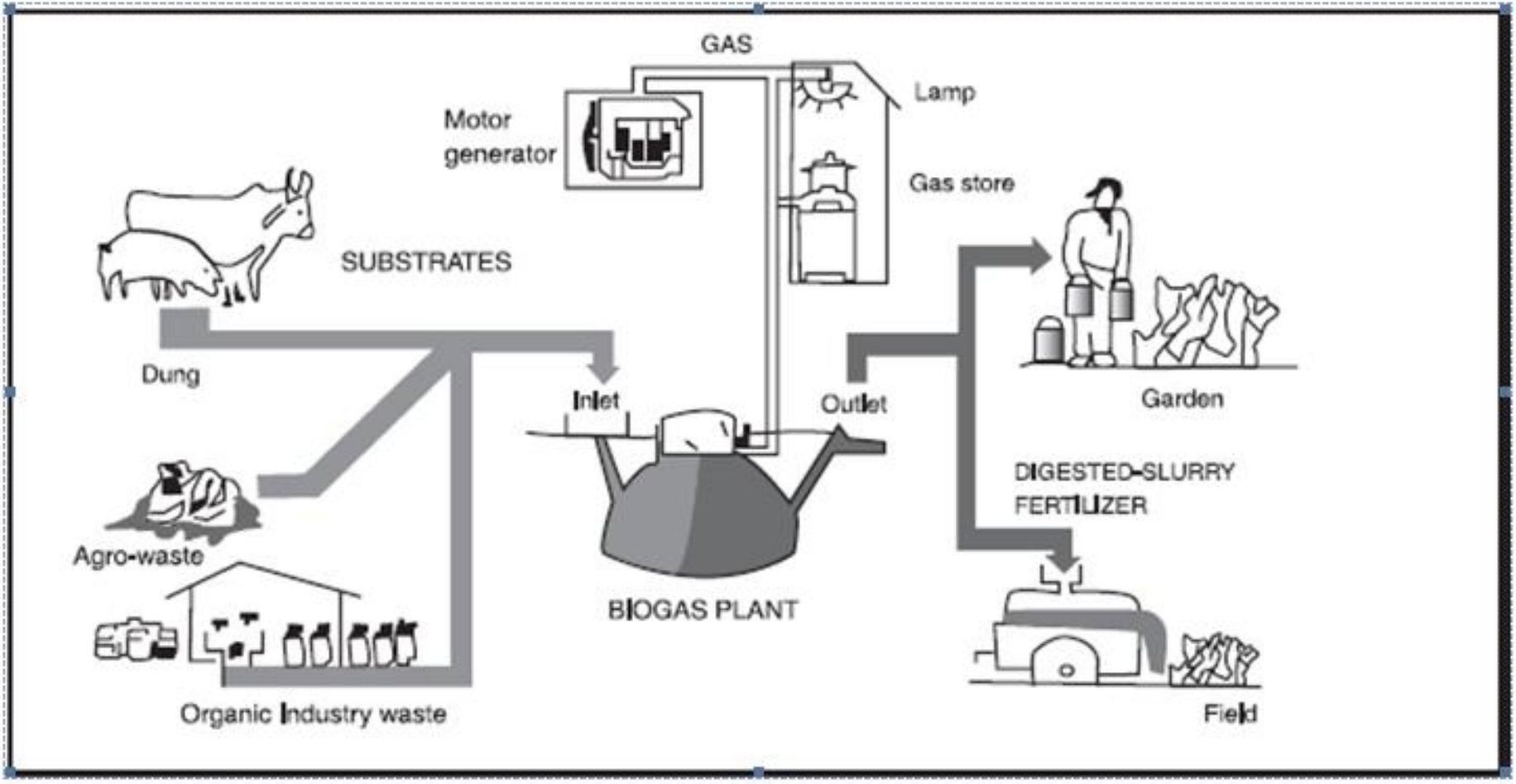
Biogas production

- Biogas is produced by bacteria during the digestion or fermentation of organic matter under airless conditions (an anaerobic process).
- The resulting gas consists typically of 60 per cent methane and 40 per cent carbon dioxide
- There are small proportions of other substances, including up to 1 per cent hydrogen sulphide
- The process itself takes place in an enclosed environment, known as biogas ‘digester’ and is called ‘anaerobic digestion’.

Flow diagram in biogas production



Schematic diagram of feedstock and unit operations in a typical biogas plant



Anaerobic Digestion (AD)

- AD is a biochemical process during which complex organic matter is decomposed in absence of oxygen, by various types of anaerobic microorganisms.
- The process of AD is common to many natural environments such as the stomach of ruminants
- In a biogas installation, the result of the AD process is the *biogas and the digestate.*
- *If the substrate for AD is a homogenous mixture of two or more feedstock types* (e.g. animal slurries and organic wastes from food industries), the process is called “co–digestion” and is common to most biogas applications today.

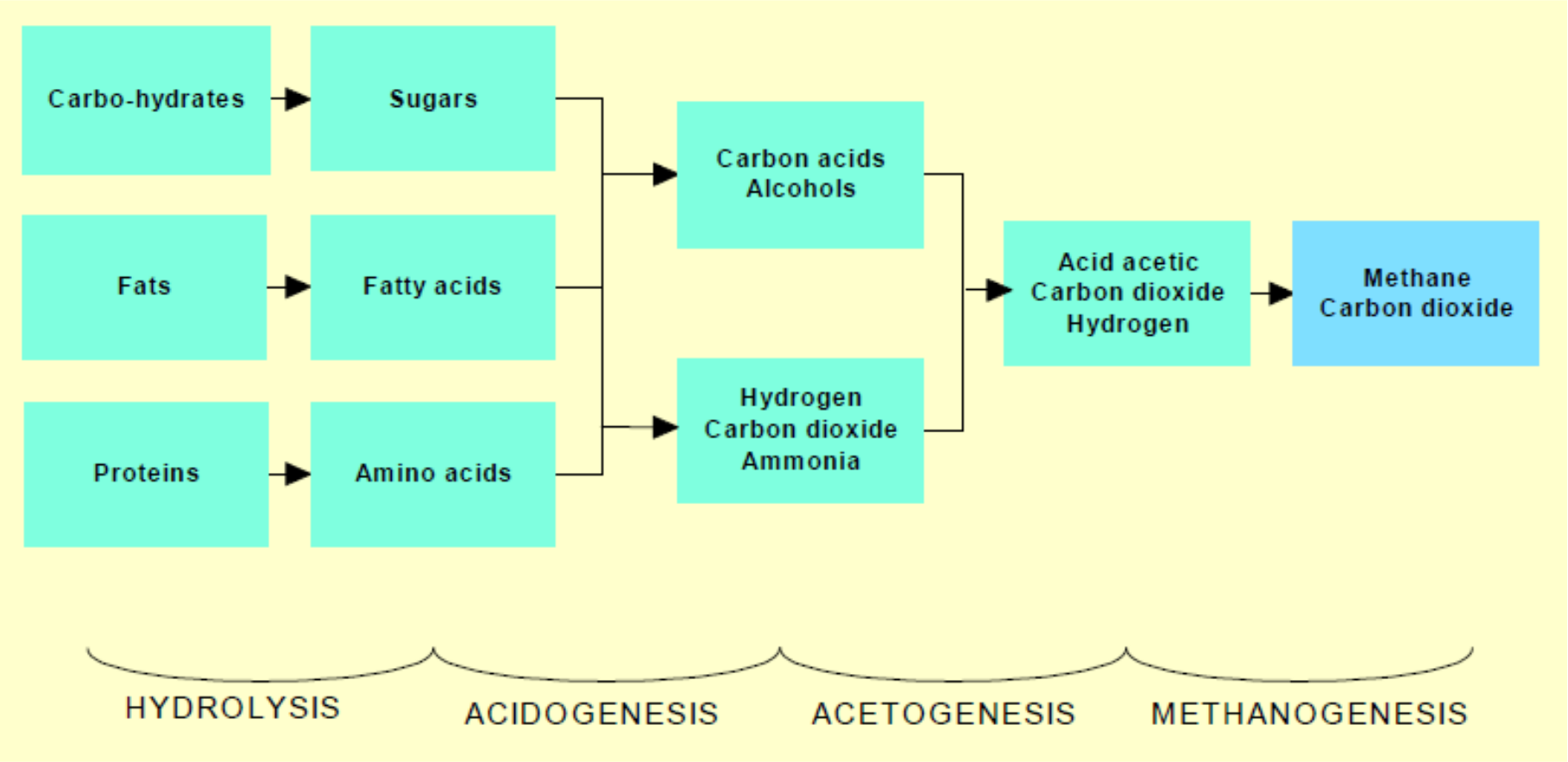
Anaerobic Digestion (AD)

- Substrates containing high amounts of lignin, cellulose and hemicelluloses can also be codigested, but a pre-treatment is usually applied in this case, in order to enhance their digestibility.
- The potential methane yield is one of the important criteria of evaluation of different AD substrates
- Animal manure has a rather low methane yield and is usually not digested alone, but mixed with other co-substrates, with high methane yield, in order to boost the biogas production.
- Common co-substrates, added for co-digestion with manure and slurries, are oily residues from food, fishing and feed industries, alcohol wastes, from brewery and sugar industries, or even specially cultivated energy crops.

The Biological Process

- The digestion process begins with bacterial hydrolysis of the input materials in order to break down insoluble organic polymers such as carbohydrates and make them available for other bacteria.
- Acidogenic bacteria then convert the sugars and amino acids into carbon dioxide, hydrogen, ammonia, and organic acids.
- Acetogenic bacteria then convert these resulting organic acids into acetic acid, along with additional ammonia, hydrogen, and carbon dioxide.
- Finally, methanogens convert these products to methane and carbon dioxide.

Stages in Anaerobic digestion



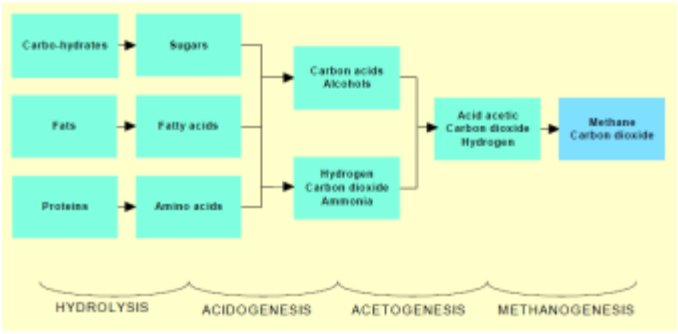
Hydrolysis

- Hydrolysis is theoretically the first step of AD, during which the complex organic matter (polymers) is decomposed into smaller units (mono- and oligomers).
- During hydrolysis, polymers like carbohydrates, lipids, nucleic acids and proteins are converted into glucose, glycerol, purines and pyridines.
- Hydrolytic microorganisms excrete hydrolytic enzymes, converting biopolymers into simpler and soluble compounds as it is shown below:

$$\text{Lipids} \xrightarrow{\text{lipase}} \text{fatty acids, glycerol}$$

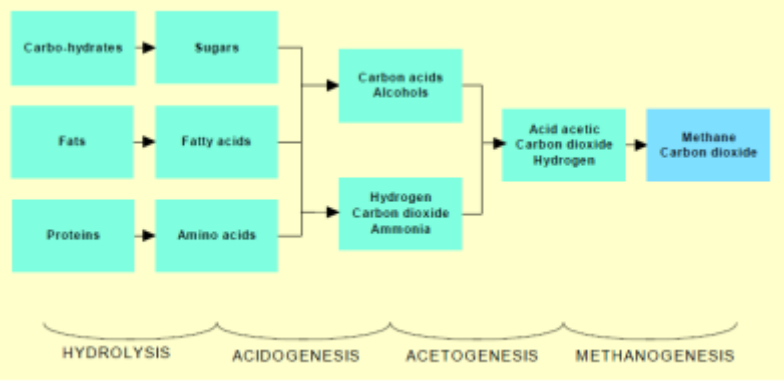
$$\text{Polysaccharide} \xrightarrow{\text{cellulase, cellobiase, xylanase, amylase}} \text{monosaccharide}$$

$$\text{Proteins} \xrightarrow{\text{protease}} \text{amino acids}$$



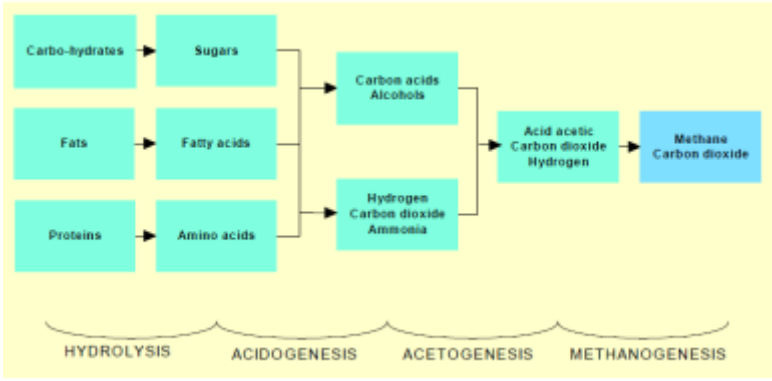
Acidogenesis

- During acidogenesis, the products of hydrolysis are converted by acidogenic (fermentative) bacteria into methanogenic substrates.
- Simple sugars, amino acids and fatty acids are degraded into acetate, carbon dioxide and hydrogen (70%) as well as into volatile fatty acids (VFA) and alcohols (30%).



Acetogenesis

- Products from acidogenesis, which can not be directly converted to methane by methanogenic bacteria, are converted into methanogenic substrates during acetogenesis.
- VFA and alcohols are oxidised into methanogenic substrates like acetate, hydrogen and carbon dioxide.
- VFA, with carbon chains longer than two units and alcohols, with carbon chains longer than one unit, are oxidized into acetate and hydrogen.

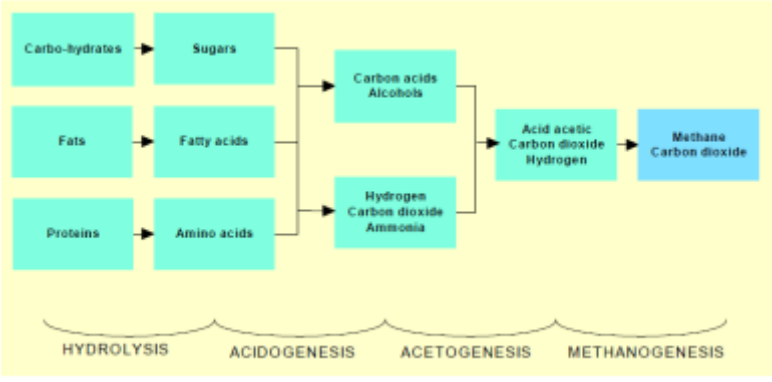


Methanogenesis

- The production of methane and carbon dioxide from intermediate products is carried out by methanogenic bacteria.
- 70% of the formed methane originates from acetate, while the remaining 30% is produced from conversion of hydrogen (H) and carbon dioxide (CO2), according to the following equations:

$$\textit{Acetic acid} \xrightarrow{\text{methanogenic bacteria}} \textit{methane} + \textit{carbon dioxide}$$

$$\textit{Hydrogen} + \textit{carbon dioxide} \xrightarrow{\text{methanogenic bacteria}} \textit{methane} + \textit{water}$$



Methanogenesis

- Methanogenesis is a critical step in the entire anaerobic digestion process, as it is the slowest biochemical reaction of the process.
- Methanogenesis is severely influenced by operation conditions. Such as *Composition of feedstock, feeding rate, temperature, and pH*
- Digester overloading, temperature changes or large entry of oxygen can result in termination of methane production.

Factors influencing anaerobic digestion process efficiency

- Temperature
- pH Value
- Exclusion of oxygen
- Nutrient supply
- Stirring intensity
- Substrate temperature
- Substrate solid content
- Hydraulic retention time (HRT)
- Organic load

Temperature

- The AD process can take place at different temperatures, divided into three temperature ranges
- The operation temperature is chosen with consideration to the feedstock used and the necessary process temperature is usually provided by floor or wall heating systems, inside the digester

Thermal stage	Process temperatures	Minimum retention time
psychrophilic	< 20 °C	70 to 80 days
mesophilic	30 to 42 °C	30 to 40 days
thermophilic	43 to 55 °C	15 to 20 days

Hydraulic retention time (HRT)

- The HRT is the average time interval when the substrate is kept inside the digester tank
- HRT is correlated to the digester volume and the volume of substrate fed per time unit, according to the following equation

$$HRT = V_R / V$$

HRT	hydraulic retention time [days]
V_R	digester volume [m^3]
V	volume of substrate fed per time unit [m^3/d]

Hydraulic retention time (HRT)

	Amount of biogas produced expressed as percentage (%) per time		
Time (Days)	Human Waste	Pig manure	Cow dung
10	40.7	46.0	34.4
20	81.5	78.1	74.6
30	94.1	93.9	86.2
40	98.2	97.5	92.7
50	98.7	99.1	97.3
60	100	100	100

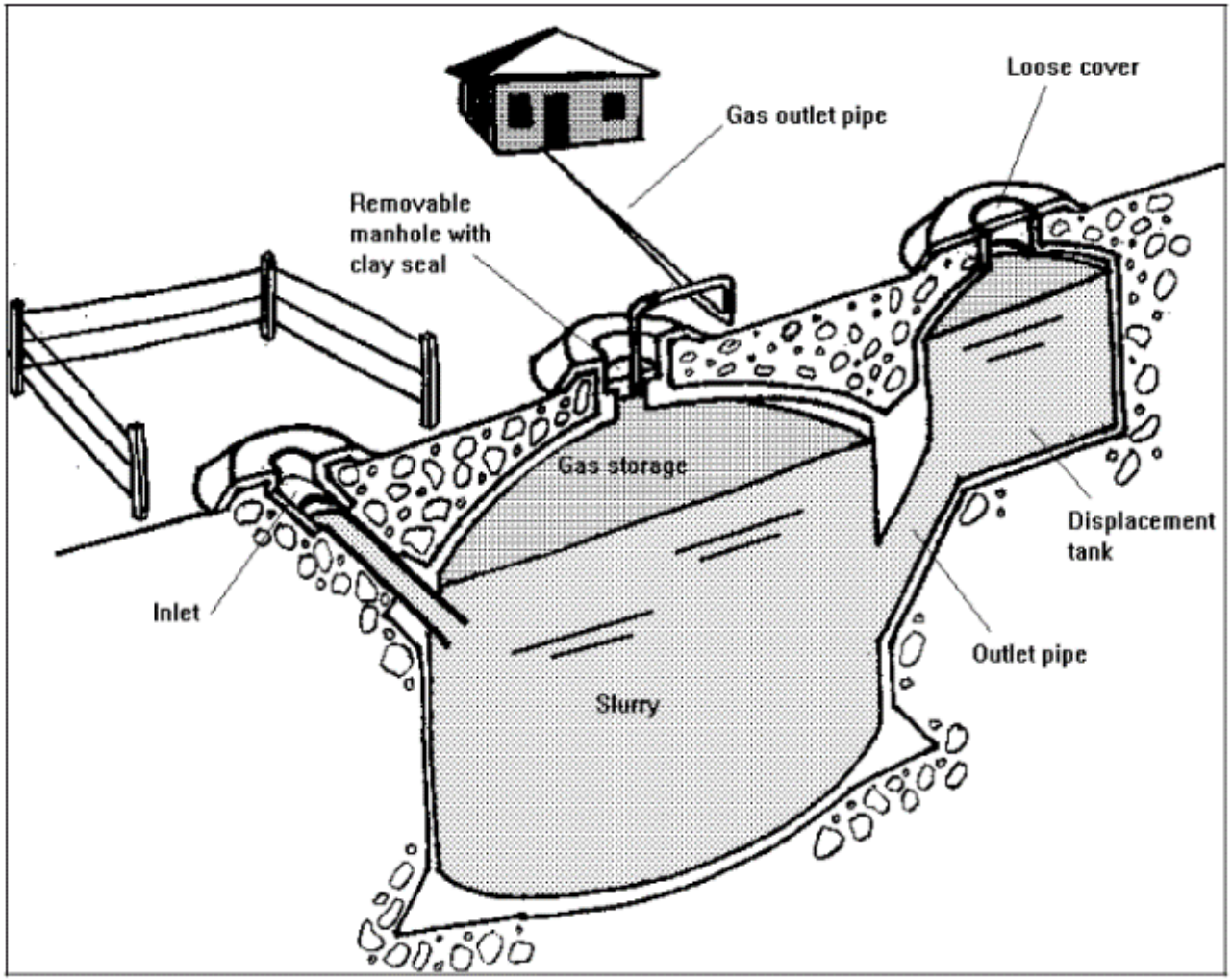
Organic load

- Indicates how much organic dry matter can be fed into the digester, per volume and time unit, according to the equation below:

$$B_R = m * c / V_R$$

B_R	organic load [kg/d*m³]
m	mass of substrate fed per time unit [kg/d]
c	concentration of organic matter [%]
V_R	digester volume [m³]

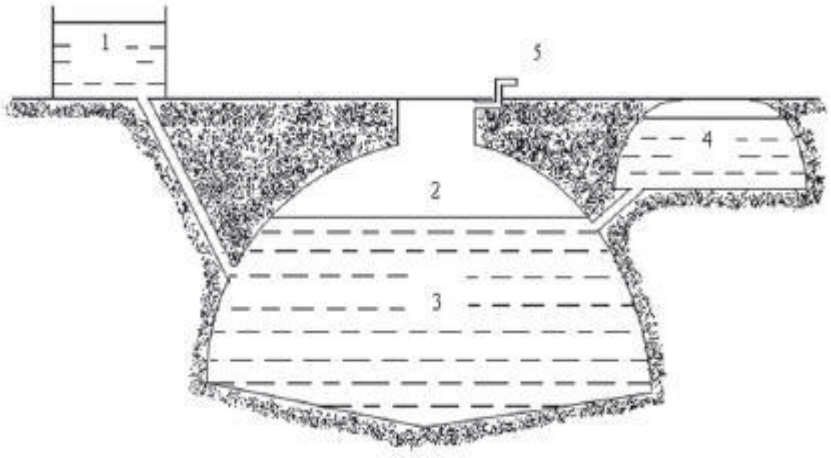
Basic concept of a digester

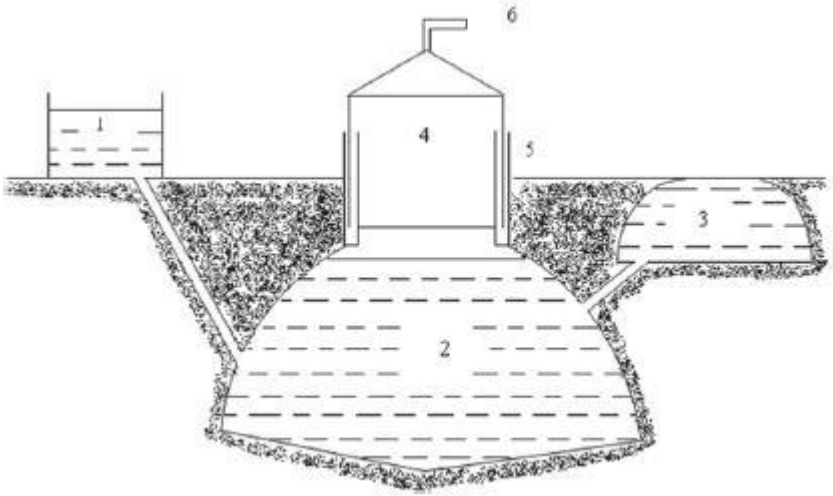


Digester designs

- Two popular simple designs of digester have been developed; the Chinese fixed dome digester and the Indian floating cover biogas digester.
- The digestion process is the same in both digesters but the gas collection method is different in each.
- Both have been designed for use with animal waste or dung.

Fixed dome and floating drum digesters compared





Fixed dome digester

1. Mixing tank with inlet pipe.
2. Gasholder.
3. Digester.
4. Compensation tank.
5. Gas pipe.

Floating drum digester

1. Mixing tank with inlet pipe.
2. Digester.
3. Compensation tank.
4. Gasholder.
5. Water jacket.
6. Gas pipe.

Advantages of fixed dome digester

- Relatively low construction costs and the absence of moving parts and of steel that can rust.
- They have a long lifespan if they are well constructed.
- Since fixed-dome plants are situated underground, space is saved and the digester is protected from temperature changes.



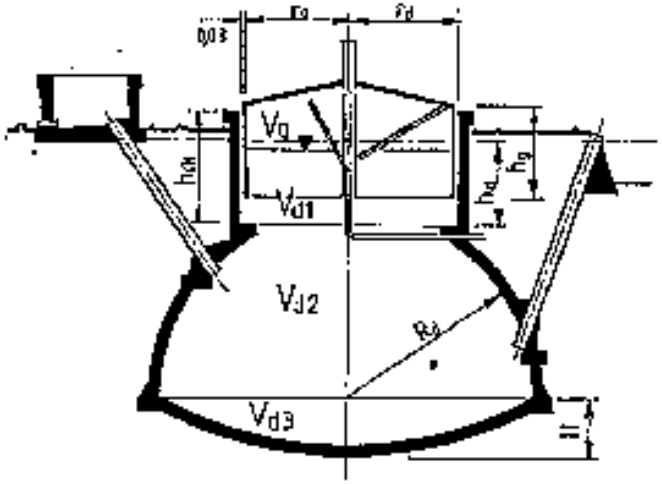


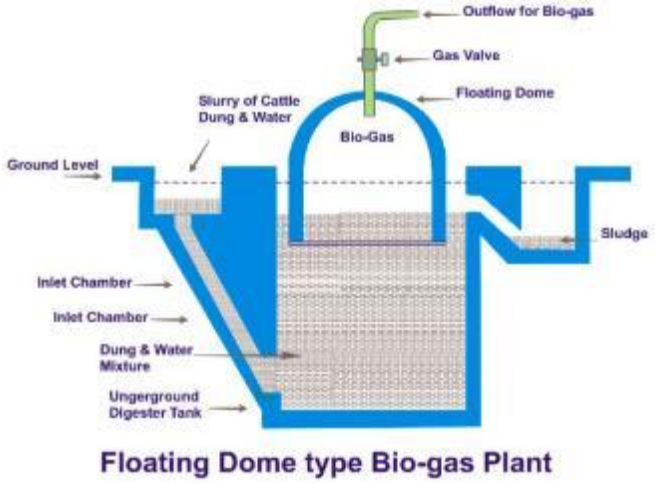
Disadvantages of fixed dome digesters

- Frequent problems with the gas-tightness of the brickwork gas holder, where even a small crack in the upper brickwork can cause a large loss of biogas.
  - Fixed-dome plants are therefore recommended only where construction can be supervised by experienced biogas technicians.
- The gas pressure fluctuates substantially depending on the volume of the stored gas.
- Even though the underground construction protects against temperature extremes, digester temperatures are low.

Advantages of floating drum digester

- The main advantages of this type of system are its simplicity and ease of operation, as the volume of stored gas is directly visible to the user.
- The gas pressure is constant and determined by the weight of the gas holder .
- Construction is relatively easy and mistakes do not lead to major problems in operation or gas yield.
- Construction can be done using local skilled labour





Floating Dome type Bio-gas Plant

Disadvantages of floating drum digester

- The disadvantages are the high material costs of the steel drum and the susceptibility of the steel parts to corrosion;
  - because of this, floating-drum plants have a shorter lifespan than fixed-dome plants
  - and there are costs associated with the regular painting of the drum.

Potentials of biogas production from some types of dung

Dung type	Potential Biogas Yield (m^3/kilo)
Cattle (cows and buffalos)	0.023 - 0.040
Pig	0.040 - 0.059
Poultry (Chicken)	0.065-0.116
Human	0.020- 0.028
Pre-treated Crop Waste	0.037
Water Hyacinth	0.045

Biogas properties

- The composition and properties of biogas varies to some degree depending on feedstock types, digestion systems, temperature, retention time, etc.
- Table below contains some average biogas composition values, found in most of the literature.

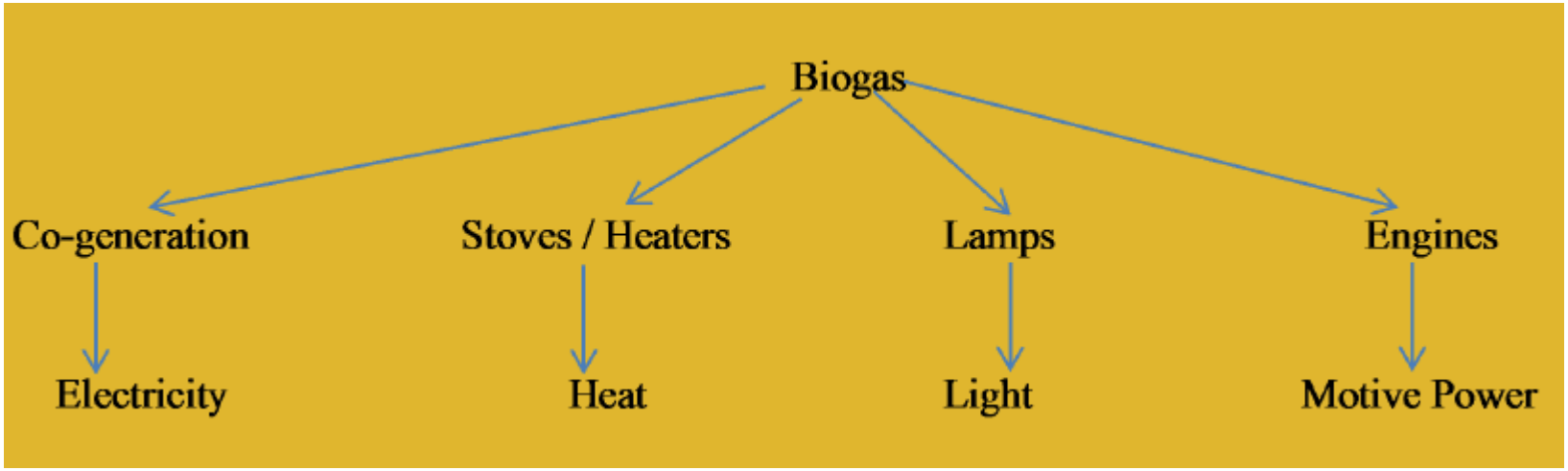
Compound	Chemical symbol	Content (Vol.-%)
Methane	CH_4	50-75
Carbon dioxide	CO_2	25-45
Water vapour	H_2O	2 (20°C) -7 (40°C)
Oxygen	O_2	<2
Nitrogen	N_2	<2
Ammonia	NH_3	<1
Hydrogen	H_2	<1
Hydrogen sulphide	H_2S	<1

Biogas application technologies

- **Agricultural biogas plants**
  - Family scale biogas plants
  - Farm-scale biogas plants
  - Centralised (joint) co-digestion plants
- **Waste water treatment plants**
- **Municipal solid waste (MSW) treatment plants**
- **Industrial biogas plants**
- **Landfill gas recovery plants**

Utilisation of biogas

- Methane is the most valuable component of the biogas used as a fuel.
- The calorific power of biogas is about 6 kWh/m^3, which corresponds to about half a litre of diesel oil; however, this value varies with altitude and temperature.



Construction of biogas plants

- The conventional material for the construction of digesters in Africa is masonry; however, digesters can be made of plastic, which is less expensive than other materials.
- Factors such as the specifications of the site, including the nature and type of soil, must be taken into account when considering structural stability.

Design

- Factors such as the specifications of the site, including the nature and type of soil, must be taken into account when considering structural stability.
- The type of feedstock to be used in the digester determines the quantity of biogas produced and therefore affects the design capacities of both the digester and the storage facility.
- The availability and choice of the construction material affect both the capital cost and the operating cost
- The availability of highly trained workers with the requisite skills affects the longevity of the plant and minimizes the need to import labour

Construction

- During the construction of digesters the following engineering issues must be taken into consideration:
  - The shape of the digester affects the susceptibility of the structure to external loading effects, especially for fixed-dome digesters, and therefore influences the structural integrity of the digester;
  - Feedstock loading affects the efficiency of the system;
  - Inlet and outlet configurations can affect the pressure dynamics of the digester by virtue of their locations, thereby affecting the operational efficiency.

Operational issues

- Slurry mixing is an essential element of the biogas system that ensures uniform feedstock and the correct slurry water content, both of which are essential for the smooth flow of the feed into the digester and hence the production of the biogas.
- Excessive water content will lead to a reduction of digestion efficiency, due to a decreased retention time, and therefore lower production of gas.

End of Lecture 6